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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Computer Science and Engineering (Cyber Security)

CRYPTOGRAPHY AND NETWORK SECURITY (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define the three primary security goals.	02	1	1
	1.2	What is the Chinese Remainder Theorem? Give example.	02	1	2
	1.3	Differentiate between stream and block ciphers.	02	1	2
	1.4	List the main transformations used in AES encryption.	02	1	2
	1.5	Name two real-world applications where block ciphers are utilized.	02	2	2
	1.6	Mention the block cipher modes of operation, and why are they important?	02	2	2
	1.7	What is the role of X.509 certificate in public key infrastructure?	02	3	2
	1.8	How does Kerberos facilitate secure authentication?	02	1	3
	1.9	How does IEEE 802.11i enhance wireless LAN security?	02	1	3
	1.10	What is the purpose of the Wireless Application Protocol (WAP)?	01	3	3
	1.11	Define IPSec policy?	01	2	4

PART-B

2	a	Explain the fundamental security goals in cryptography. Discuss various types of cryptographic attacks and the corresponding security services and mechanisms designed to counter them.	10	2	2
	b	Describe key cryptographic techniques and the mathematical foundations that support the security services and mechanisms.	06	3	4
3	a	Explain the key features of the Advanced Encryption Standard (AES), including its transformations and key expansion process.	10	2	2
	b	How does AES improve upon DES?	06	2	2
		OR			
4	a	Describe the structure and operation of the Data Encryption Standard (DES). Discuss its security vulnerabilities and the reasons for its replacement.	08	2	2
	b	Explain the principles of substitution and transposition ciphers. How do stream and block ciphers differ in their operation and applications?	08	2	2
5	a	Give short notes on i) Rabin Cryptosystem ii) Elgamal Cryptosystem	08	3	4
	b	Write the encryption algorithm pseudocode for output feedback (OFB) and Counter mode (CTR)	08	2	4

		OR			
6	a	Explain why there is no need for cipher text stealing in CFB, OFB and CTR	06	2	2
	b	Explain the RSA algorithm in detail, including the key generation, encryption, and decryption processes. Discuss the mathematical principles that ensure its security.	10	2	2
7	a	Explain the requirements for message authentication and describe how Message Authentication Codes (MACs) fulfill these requirements.	08	2	2
	b	Compare and contrast the MD5 and SHA-3 cryptographic hash functions in terms of structure, security, and applications.	08	2	2
		OR			
8	a	Write 2 algorithms for Elgamal scheme: one for the signing process and one for the verifying process.	08	4	3
	b	Using the RSA scheme, let p=809, q=751, and d=23. Calculate the public key e. Then i) Sign and verify a message with M1=100.Call the signature S1. ii) Sign and verify a message with M2=50.Call the signature S1. iii) Show that if M=M1 X M2=5000, then S=S1XS2	08	4	3
9	a	Explain the role of Transport Layer Security (TLS) in web security. How does TLS enhance secure communication over the internet?	08	4	3
	b	What is IP Security (IPSec), and how does the Encapsulating Security Payload (ESP) component contribute to secure IP communications?	08	2	3
		OR			
10	a	Discuss the security enhancements introduced by IEEE802.11i in wireless LANs. How does it improve upon previous standards?	08	1	4
	b	Explain the concept of WPA3 and how it improves upon WPA2 in securing wireless networks. Include a discussion on Simultaneous Authentication of Equals (SAE).	08	2	3